

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Designed a dual-database layer that keeps transactional records in PostgreSQL while serving relationship-heavy reads from Neo4j, routing each query class to the store that answers it fastest instead of forcing one engine to do both jobs.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent, sustaining 150+ check-ins per event.
- Tuned React Query cache tiers by data volatility across the dashboards, caching stable roles longer than fast-changing application queues to cut redundant refetches while keeping live data fresh.

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.
- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.

PROJECTS

Dadei | Python, FastAPI, Redis, React

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.
- Root-caused the worker queue's silent exit-zero death on a Redis drop, then added a health check that detects it and a three-layer restart that recovers it, running the Python backend as two services from one Docker image.
- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Split speech recognition into a Deepgram streaming path for live command latency and a batched Whisper path for ambient bulk transcription, keeping interactive commands responsive while bulk transcription stays cheap.

fliks | Go, PostgreSQL, AWS EC2

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box.
- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.

TECHNICAL SKILLS

Languages Python, C++, Go, SQL, JavaScript/TypeScript
Frameworks FastAPI, React
Databases PostgreSQL, Neo4j, Redis
Cloud & Infrastructure AWS EC2, Docker
Tools Git, Linux